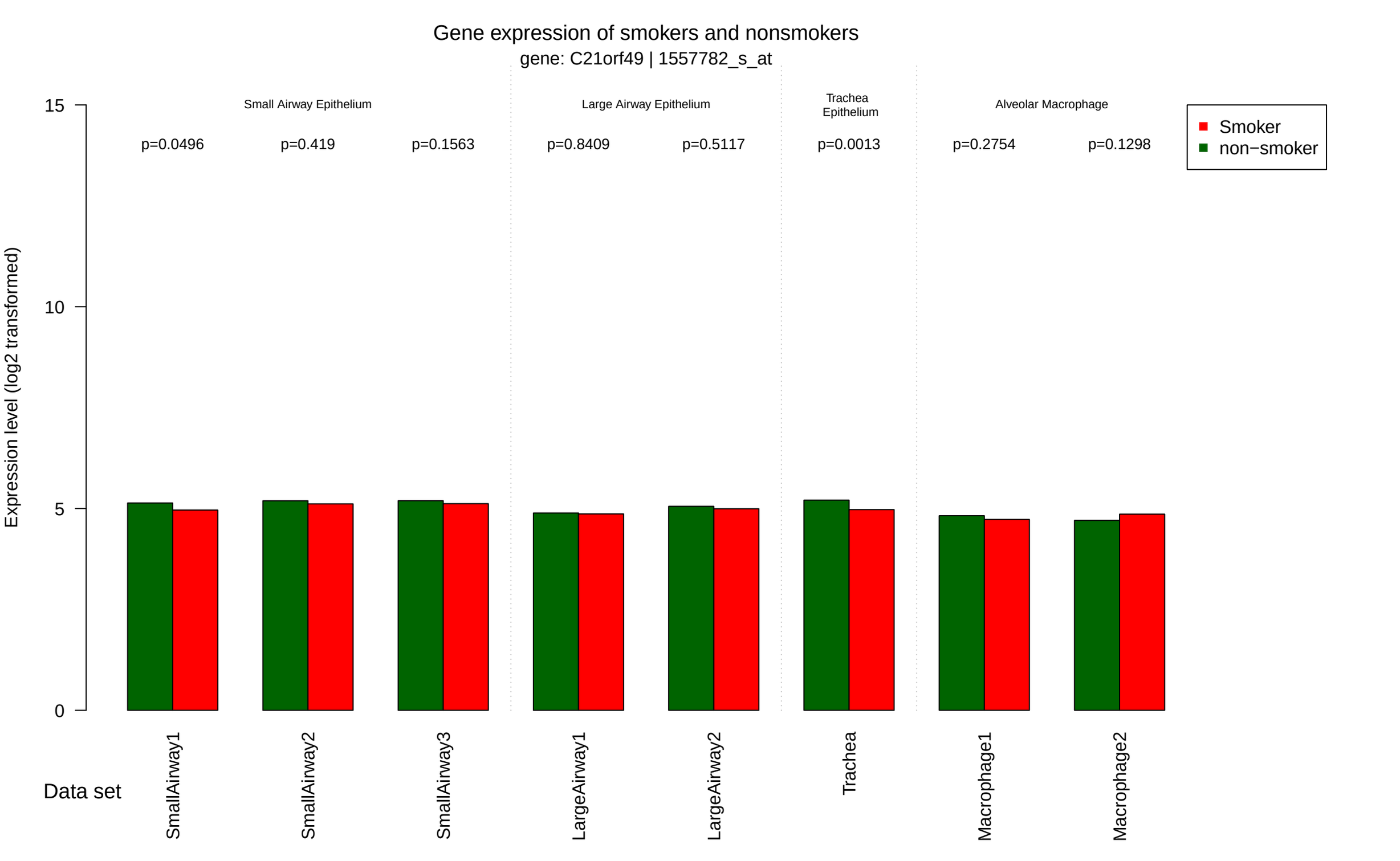
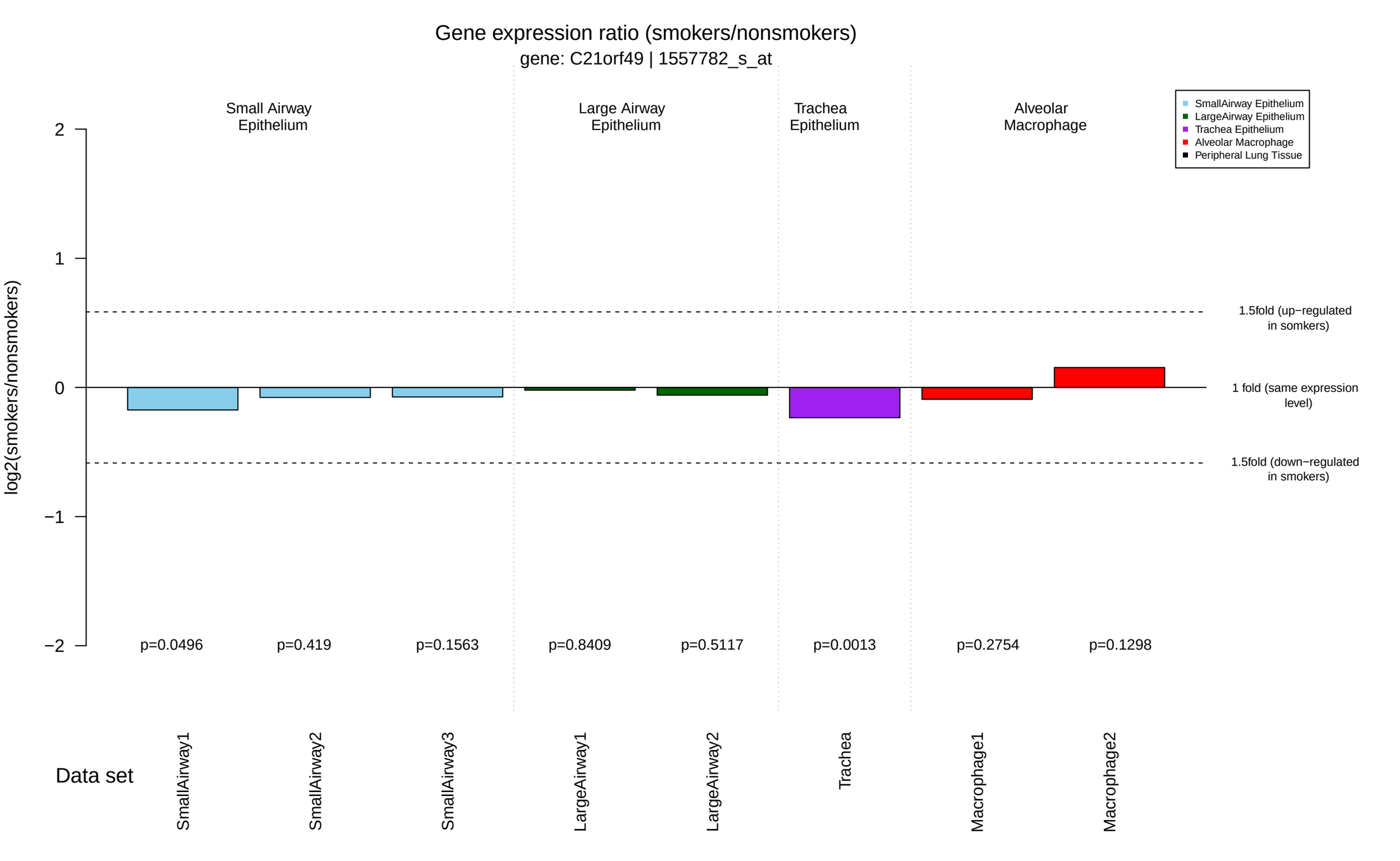
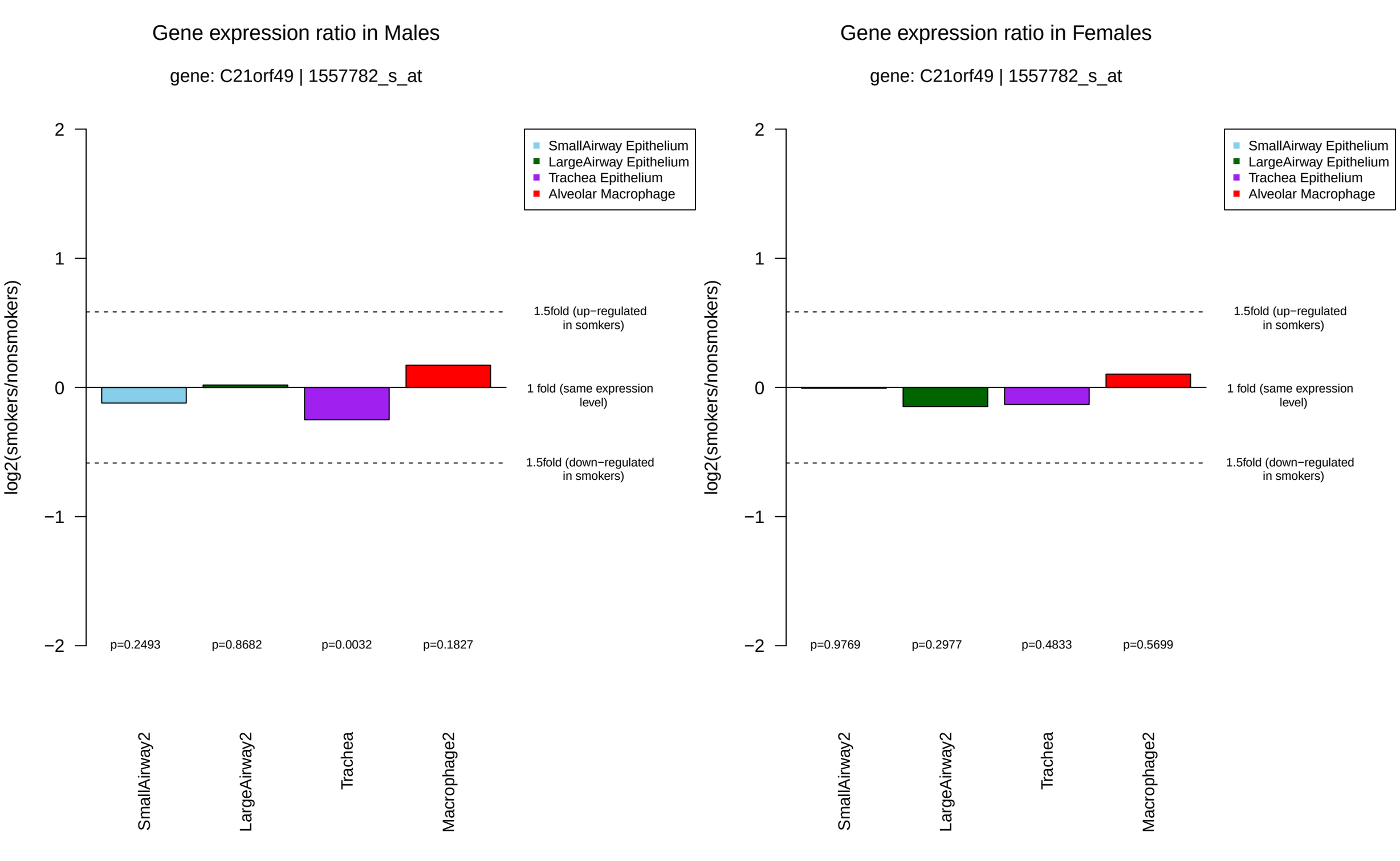
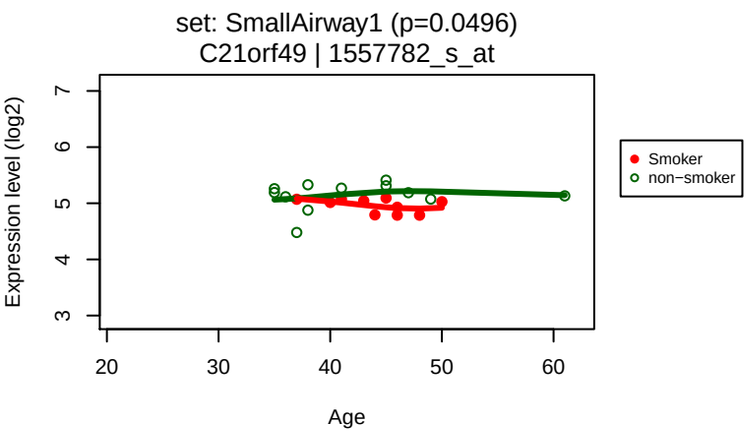
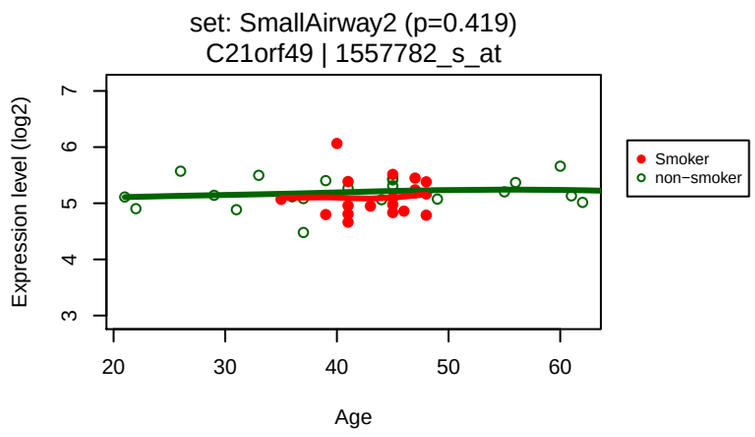
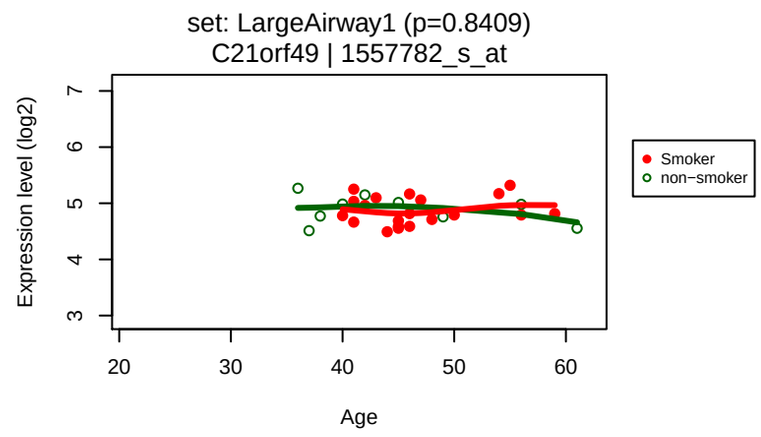

Gene expression of smokers and nonsmokers
gene: C21orf49 | 1557782_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0496
p=0.419
p=0.1563
p=0.8409
p=0.5117
p=0.0013
p=0.2754
p=0.1298
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

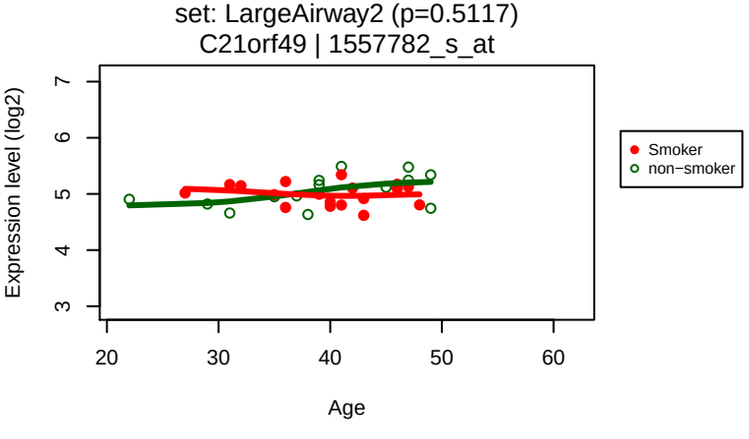

Gene expression ratio (smokers/nonsmokers)
gene: C21orf49 | 1557782_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0496
p=0.419
p=0.1563
p=0.8409
p=0.5117
p=0.0013
p=0.2754
p=0.1298
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

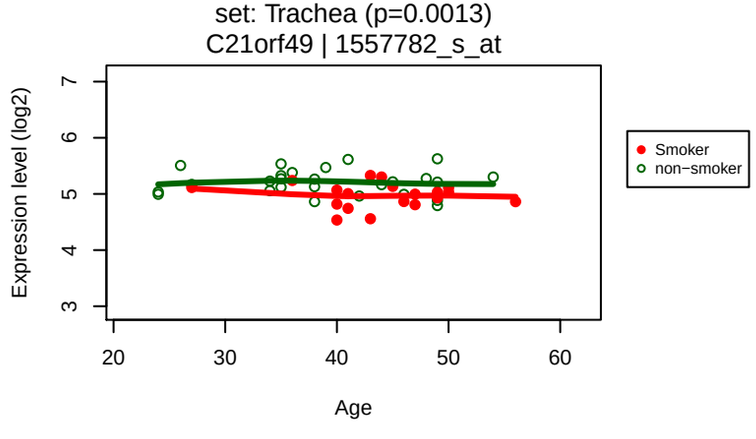

Gene expression ratio in Males
gene: C21orf49 | 1557782_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2493
p=0.8682
p=0.0032
p=0.1827
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: C21orf49 | 1557782_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9769
p=0.2977
p=0.4833
p=0.5699
SmallAirway2
LargeAirway2
Trachea
Macrophage2

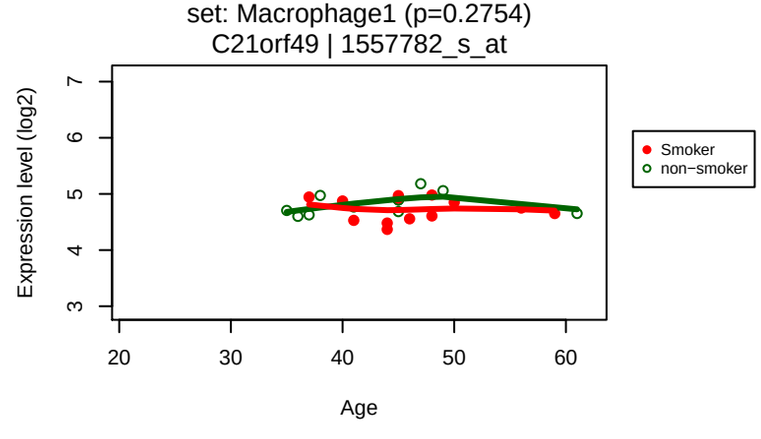

set: SmallAirway1 (p=0.0496)
C21orf49 | 1557782_s_at
Expression level (log2)
Age
Smoker
non−smoker

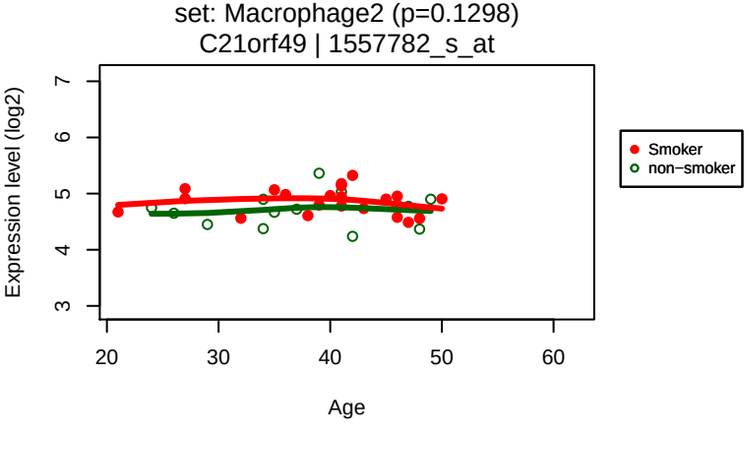

set: SmallAirway2 (p=0.419)
C21orf49 | 1557782_s_at
Expression level (log2)
Age
Smoker
non−smoker

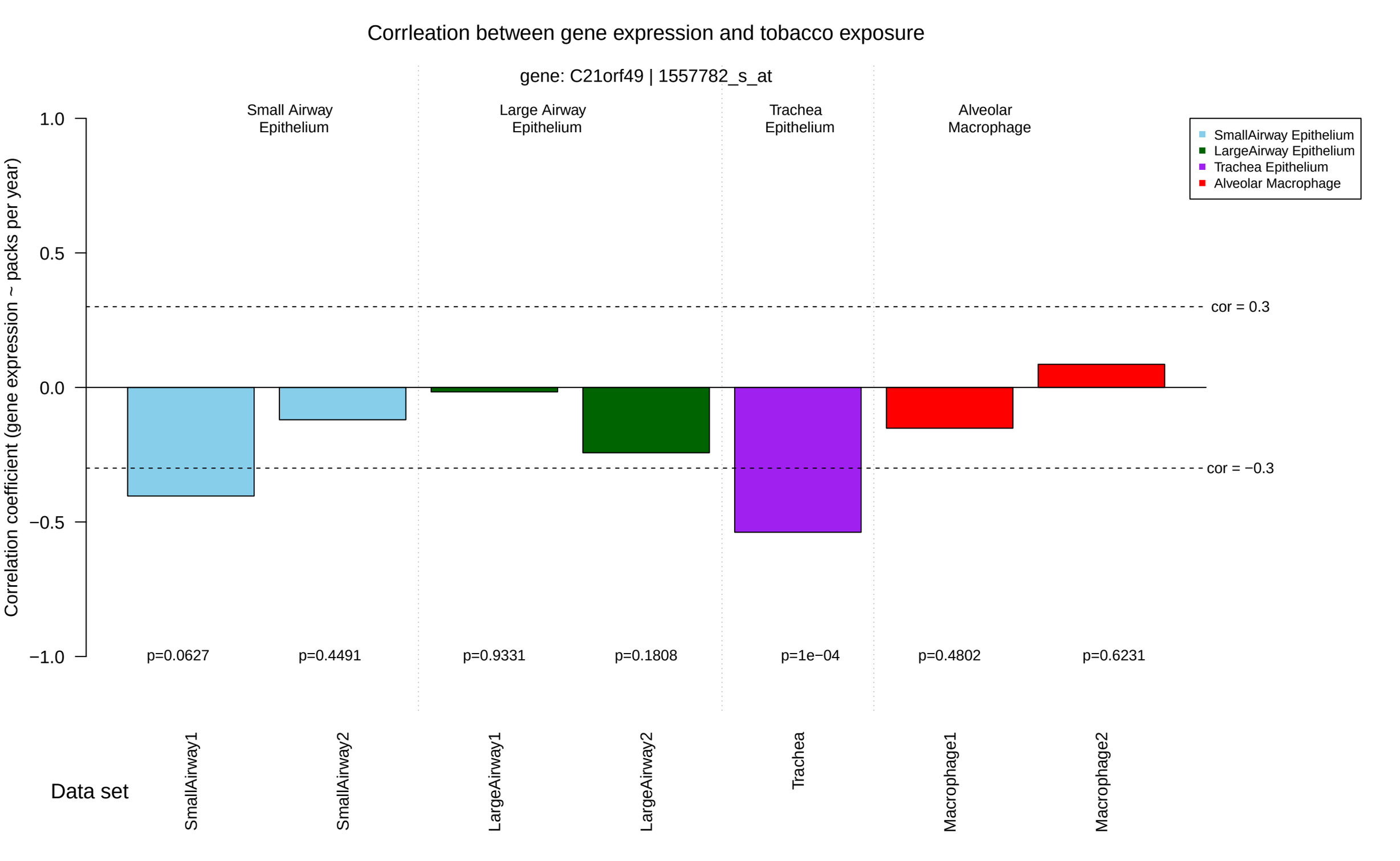

set: LargeAirway1 (p=0.8409)
C21orf49 | 1557782_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5117)
C21orf49 | 1557782_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0013)
C21orf49 | 1557782_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2754)
C21orf49 | 1557782_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1298)
C21orf49 | 1557782_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: C21orf49 | 1557782_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0627
p=0.4491
p=0.9331
p=0.1808
p=1e−04
p=0.4802
p=0.6231
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2